Experiment No:6

The microstructure refinement and the influence on the properties

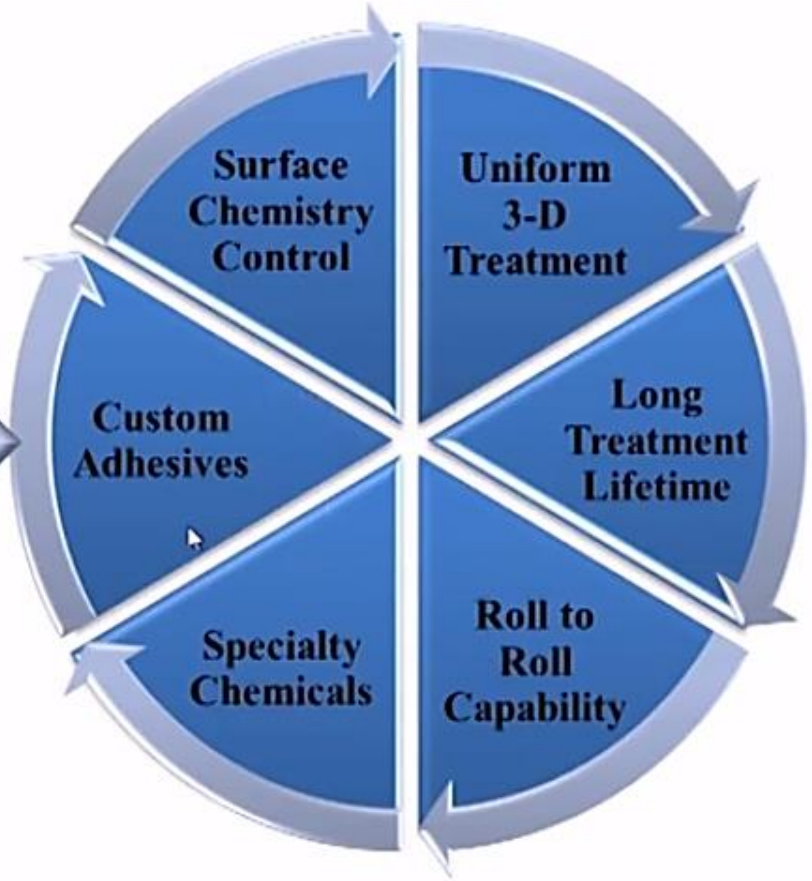
Surface Modification:

- It is an act of modifying the surface of a material by physical, chemical or biological characteristics different from the ones originally found on the surface of a material.

Purposes of surface modification

- Enhancing anticorrosive properties.
- Improving electrical conductivities and optical properties.
- Increasing bioactivities.
- Improving adhesion properties.
- Improves component lifetime and resistance to aggressive environments.
- Produce biocompatibility.
- Create permanent wettability.
- Produce hydrophobic characteristics.

Some of the benefits include:



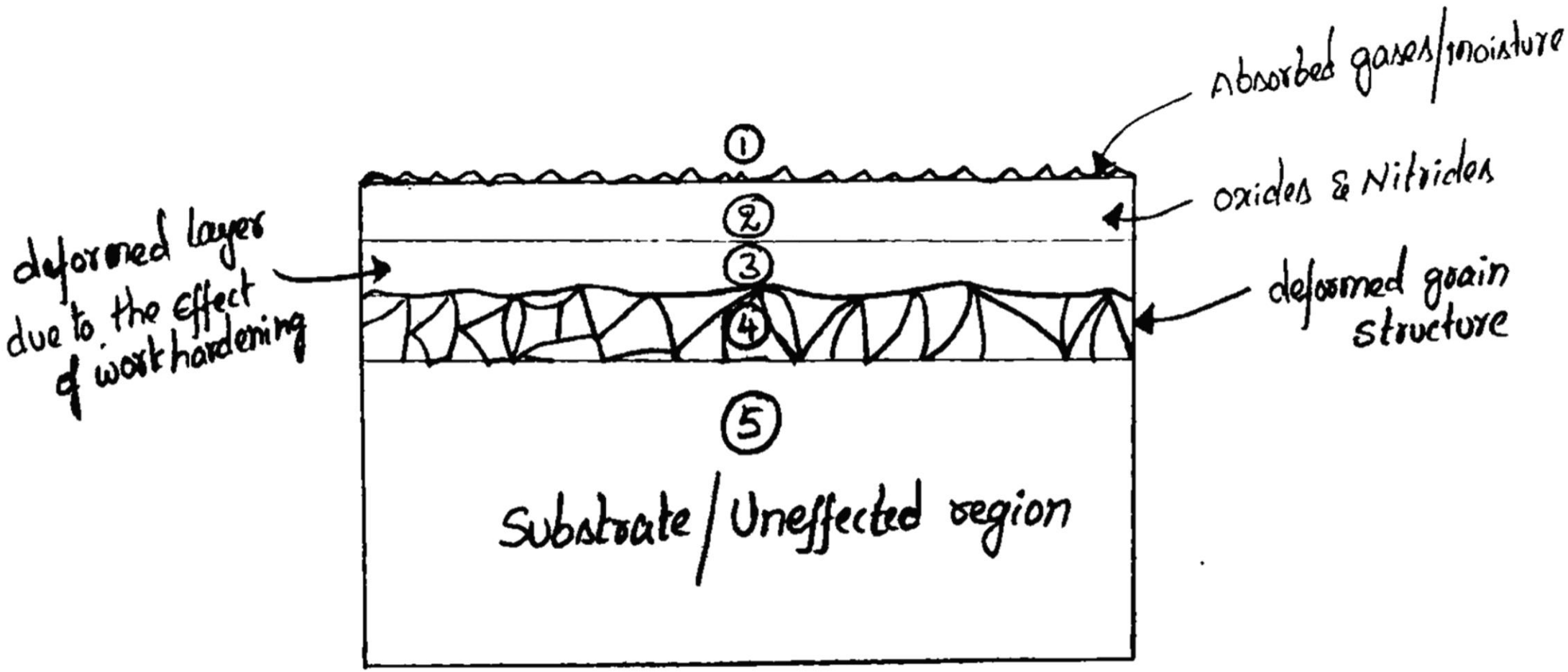
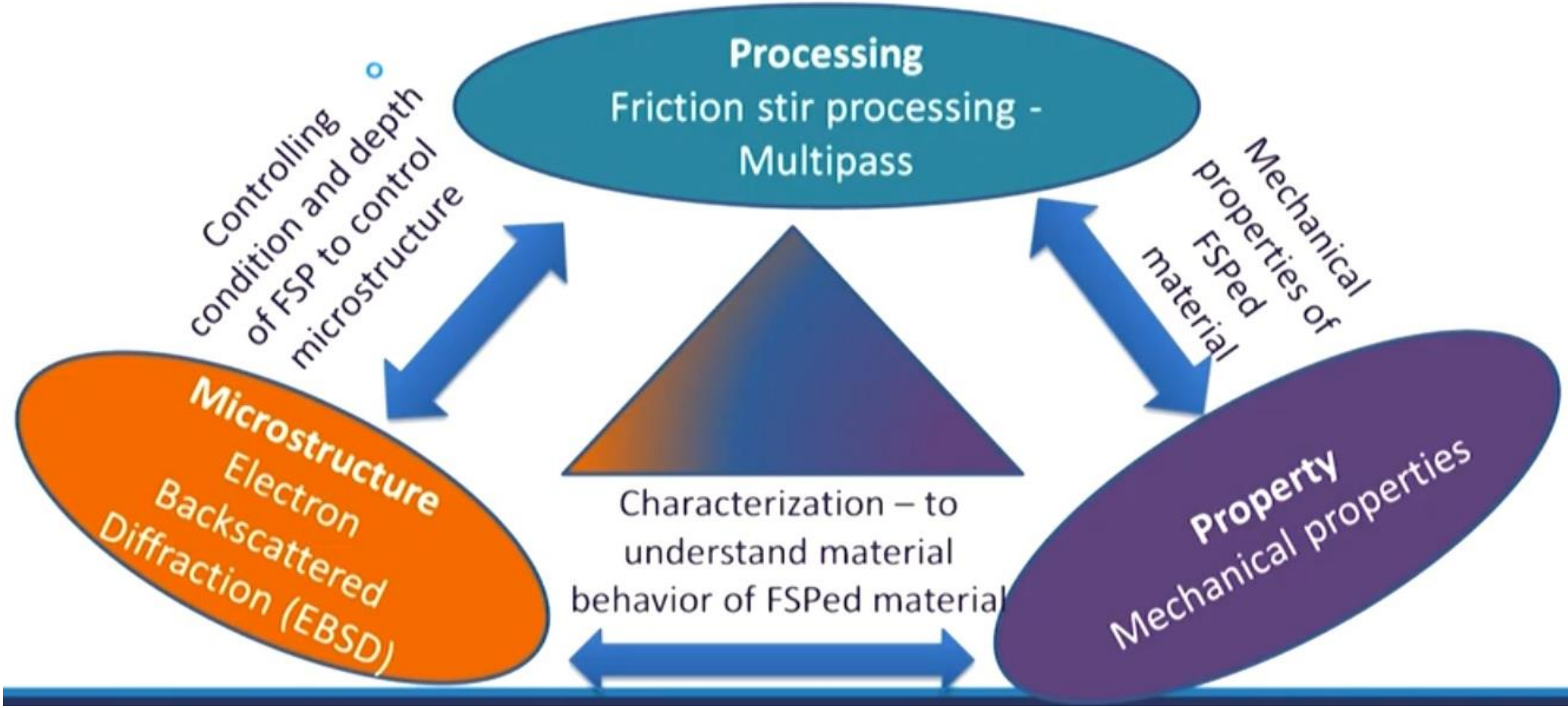


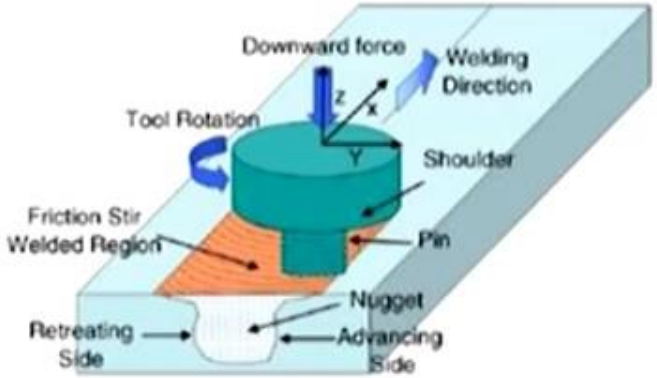
Fig: schematic of sub-surface roughness

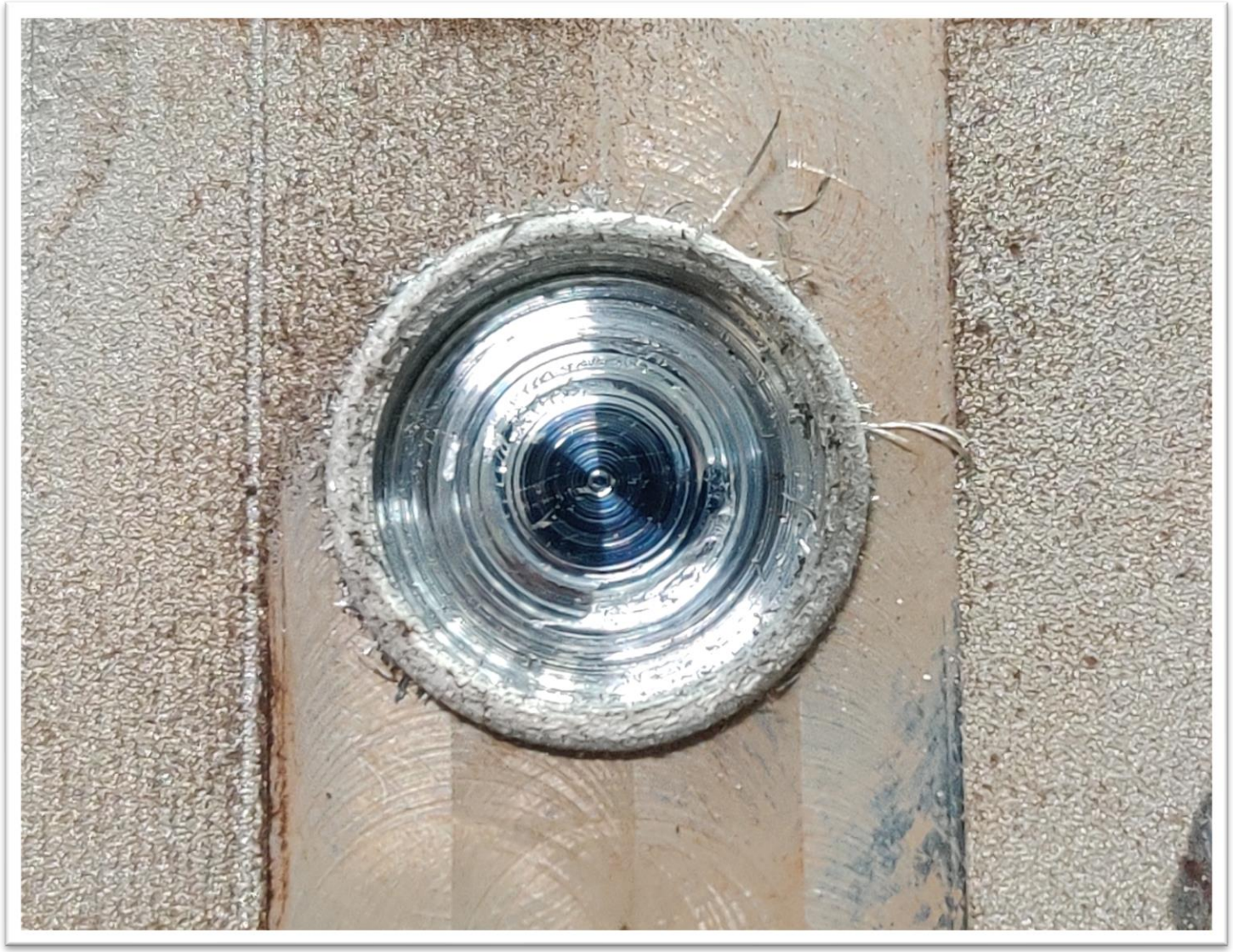
Processing – Microstructure - Property

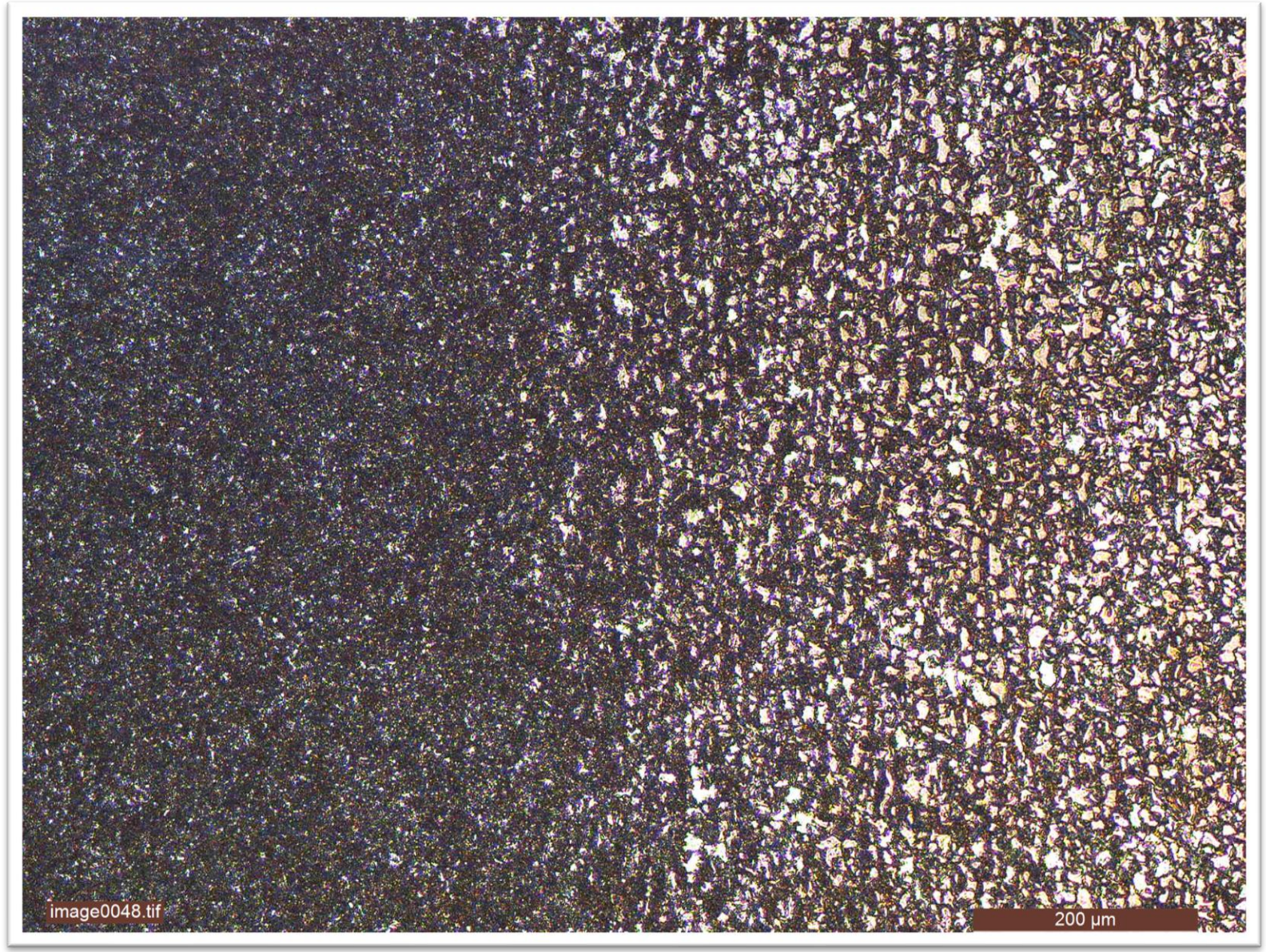


Introduction

- FSP is a derivative of friction stir welding technique
- A rotating tool consist of a shoulder and protruding pin is inserted in the material.
- Friction between surface of the sample and the shoulder generate heat which result in softening of the material
- The inserted pin stir the material around it.



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